Exercise 3

Consider two-neutrino oscillation with

— $E = 1\,\mathrm{GeV}$

— $\sin^2 2\theta = 1$

— $\Delta m^2 = 2.5 \cdot 10^{-3}\mathrm{eV}^2$

Plot $P(\nu_\mu \to \nu_\tau)$ as a function of distance/(Earth diameter).

Exercise 4

The cross-sections for neutrino and anti-neutrino scattering are in general very different.

— Does it mean that neutrino is a Dirac particle?

—If not why different cross-section do not imply Dirac nature?